Linear Programming

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Linear Programming

INTRODUCTION:

The mathematical definition of linear programming is optimizing a linear function with some constraints in linear inequalities form.

The term "linear programming" consists of two words a linear and programming. The term "linear" describes the relationship between multiple variables with first degree. The term "programming" refers to the process of choosing the best solution for a variety of alternatives

Linear Programming model are used in resource development. It can be used in manufacturing, calculating how to allocate staff and equipment to reduce operating costs.

It can be used in high-quality business activities, to determine what products should be sold and at what prices to maximize the profits. It can also be used in transportation, to determine how resources are used to get work done in a limited amount of time.

In this chapter, we will solve problems of linear programming by graphical method only, although there are many other ways to solve those problems.

MATHEMATICAL MODELING:

Let understand the mathematical formulation by a simple example of the manufacturing industry.

The industry produces chairs and tables. Cost and time require to manufactured a chair is ₹ 2 and 3 hours and for table ₹ 4 and 2 hours respectively. The factory can operate 150 hours maximum with budget of ₹ 220 to produce these products per week. If each chair sells for ₹ 6 and each table sells for ₹ 7, then how many individual products should be made this week to increase profits?

Let x be the number of chairs produced, and let y be the number of the table produced. Then the first constrain in form of inequality is

$$2x + 4y \leq 220 \qquad \ldots(1)$$

Time constraints

$$3x + 2y \leq 150 \qquad \ldots(2)$$

Further, two very common constraints are written below as It's not possible to produce less than 0 of any components, so constraints are

$$\begin{cases} x \geq 0 \\ y \geq 0 \end{cases} \quad \ldots(3)$$

These are called non-negative constraints.

All 3 equations of the above form a system of constraints.

$$\begin{cases} 2x + 4y \leq 220 \\ 3x + 2y \leq 150 \\ x \geq 0 \\ y \geq 0 \end{cases} \quad \ldots(4)$$

Each chair costs ₹ 2 to make and sells for ₹ 6. This gives a profit of ₹ 4 per chair. Each table costs ₹ 4 to make and sells for ₹ 7. This gives a profit of ₹ 3 per table. The profit function can be defined as

profit $Z = 4x + 3y$ ...(5)

IMPORTANT TERMINOLOGIES USED IN LINEAR PROGRAMMING:

With the help of the above example, we will understand key definitions used in Linear Programming.

(a) Decision Variables:

The total number of chair and table units specified in x & y respectively are my decision variables. Decision variables are under our control.

(b) Objective Function:

The function which we need to maximize or minimize to get the optimal value. Z is our objective function(**equation-5**) as the company wants to maximize the profit.

(c) Constraints:

The linear inequalities on the decision variables of LPP are called constraints. In the above example, **equation (4)** contains all constraints.

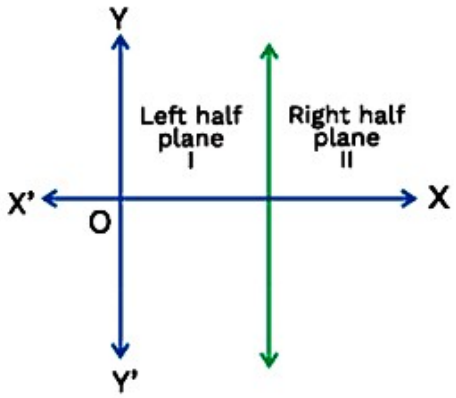
(d) Non-Negative constraints:

In linear programming decision variables x and y are taken as non-negative integers **(equation-3).**

GRAPHICAL SOLUTION OF LINEAR INEQUALITIES IN TWO VARIABLES:

A line divides the XY plane into two parts and each part is called half-plane as shown in the figure.

A point within the XY- plane can either lie on a line or can be either of the half-planes I or II.



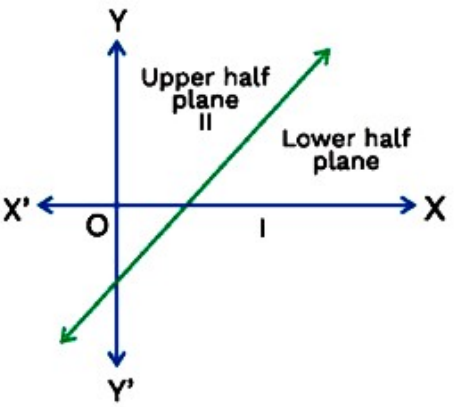
Steps to find solution region :

(i) Solution region in the xy plane is a collection of points that satisfies the inequality

(ii) In order to find the solution region, it is advised to take any point (generally (0, 0)) and check whether it satisfies the inequality or not.

(iii) For inequality $Ax + By + C \geq 0$ or ≤ 0, solid line $AX + BY + C = 0$ are also the part of solution region.

(iv) For $Ax + By + C > 0$ or < 0, dotted line $AX + BY + C = 0$ are not part of solution region.



Q. **Solve $4x + 2y > 8$ graphically**

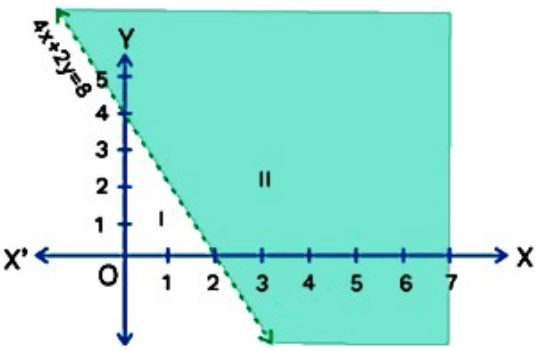
Sol. **Step-1:** Draw the dotted line $4x + 2y = 8$ in the XY plane.

Step-2: Choose a point generally (0,0) which does not lie on a line and put in given inequality

$0 > 8$ which is false

So, the region (I), where (0,0) lies, does not represent the solution region.

Hence, region (II) excluding the points on the line represents the solution region.



Q. **Solve 40x – 20y ≤ 120 graphically**

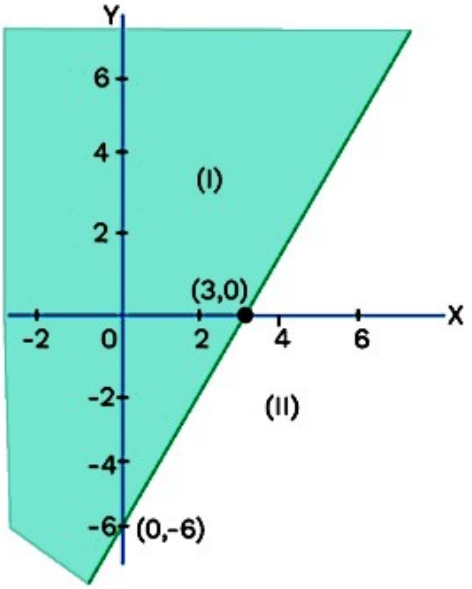
Sol. **Step-1:** Draw the Solid line

40x – 20y = 120 in the XY plane.

Step-2: Choose a point generally (0,0) which does not lie on a line and put in given inequality

0 ≤ 120 which is right

So, the region (I), where (0,0) lies, represents the solution region.



SOLUTION OF SYSTEM OF LINEAR INEQUALITIES IN TWO VARIABLES

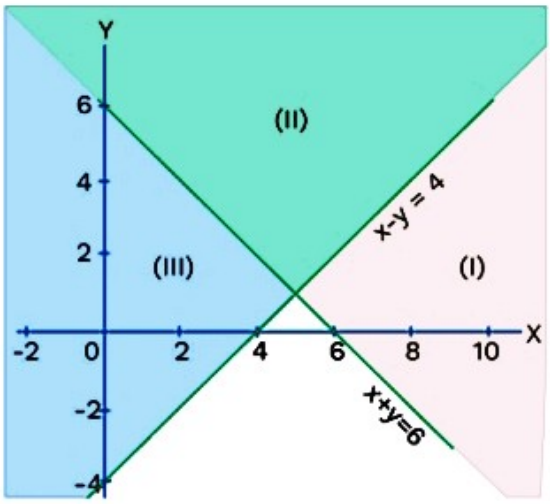
Q. **Represent solution region of linear inequalities on XY plane**
x + y ≥ 6; **x – y ≤ 4**

Sol. x + y ≥ 6 ...(1)
x – y ≤ 4 ...(2)

Two equations divide the XY plane into four regions.

Step-1: First draw the solution region of equation (1) and equation (2) as shown in the figure.

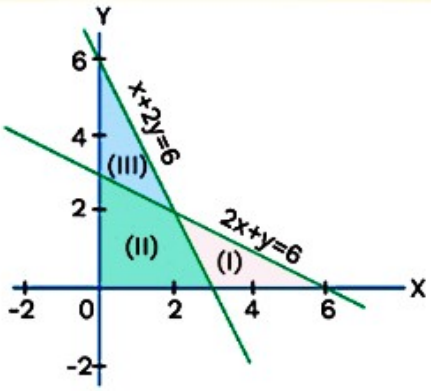
Step-2: The solution region of inquality (1) and inquality (2) is represented by region (I), (II) and (II), (III) respectively. Common region II is a required solution region of a given system of inequality.



Q. **Represent solution region of linear inequalities on XY plane**
$x + 2y \leq 6$; $2x + y \leq 6$; $x \geq 0$; $y \geq 0$

Sol.

$x + 2y \leq 6$ ...(1)
$2x + y \leq 6$ ...(2)
$x \geq 0$ ...(3)
$y \geq 0$ ...(4)



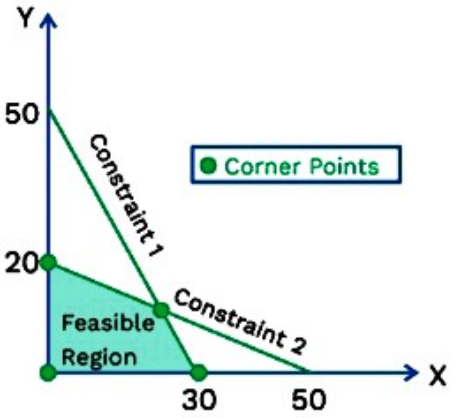
Since $x \geq 0$, $y \geq 0$, the solution region will lie in the first quadrant only.

The solution region of inequality (1) and (2), represented by region (I), (II) and (II), (III) respectively, lies below the lines.

Solution region is represented by region (II).

ANALYSIS OF SOLUTION REGION BY THE GRAPHICAL METHOD IN THE LPP MODEL

The collection of points that satisfies all of the constraints is known as a **feasible solution** and that region in XY plane is a **feasible region** and region in which feasible solution does not lie is known as infeasible region.



Fundamental Theorem

Theorem-1: Let S and Z = Ax + By be the feasible region and the objective function respectively for an LPP (Linear Programming Problem). When objective function Z has an optimal solution, then this optimal value always lies on corner points.

Theorem-2: Let S and Z = Ax + By be the feasible region and the objective function respectively for an LPP (Linear Programming Problem). For bounded region S, maximum and minimum values of Z generally exist at corner points of the feasible region.

STEPS TO FOLLOW FOR THE SOLUTION OF THE LPP MODEL :

Steps	Bounded Region	Unbounded Region
1	Determine vertex points of the feasible region	Determine vertex points of the feasible region
2	Evaluate objective function Z at corner points	Evaluate objective function Z at corner points
3	Let M is maximum and m is the minimum value of Z	Let M is maximum and m is the minimum value of Z
4	Max. of Z is M and min. of Z is m	M is the maximum value, If Ax + By > M does not pass through the feasible region. Otherwise, no maximum value exists
5		m is the minimum value, if Ax + By < m does not pass through the feasible region. Otherwise, no minimum value exists.

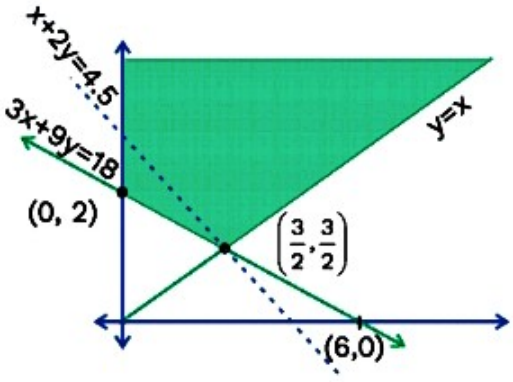
Note: This method is also known as **Corner Point Method.**

Q. **Solve the following linear programming problem graphically.**
maximize z = x + 2y
subject to constraints
3x + 9y ≥ 18
y – x ≥ 0
x ≥ 0 and y ≥ 0

Sol. The solution region of constraints are shaded as shown in the figure.

Now evaluate the value of Z at vertex points (0,2) and (1.5,1.5)

Corner points	z = x + 2y
(0,2)	4
(1.5,1.5)	4.5 = max



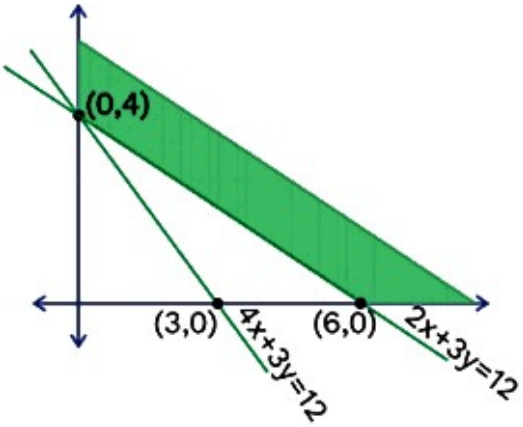
As solution region is un-bounded we need to draw the x + 2y > 4.5 and feasible region have common points with solution region of In-equality x + 2y > 4.5
Hence no max value of z exist.

Q. **Minimize z = x + 1.5y**
subject to constraints
2x + 3y ≥ 12 ; 4x + 3y ≥ 12
find the optimal solution of objective function.

Sol. The solution region determined by the given constraints is un-bounded and corner-points are found by solving linear-equations.
Now evaluate the value of z at corner points.

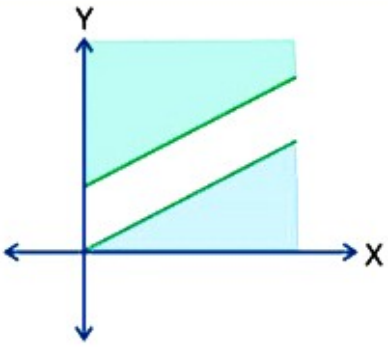
Corner points	z = x + 1.5y
(6,0)	6 = min
(0,4)	6 = min

The value of z has the same value at both corner-points.
Hence, all points lying on the line segment joining (0, 4) and (6, 0) are optimal solution.



Q. **Solve the linear programming problem by graphical method :**
maximize z = 4x + 5y, If possible subject to constraints
x – y ≤ –2 ; – x + y ≤ 0 ; x ≥ 0 ; y ≥ 0

Sol. Draw the solution region of x – y ≤ –2 and – x + y ≤ 0 in first quadrant.
Solution region of x – y ≤ – 2 is represented by green colour and of –x + y ≤ 0 by light Blue colour.
As observed by graph, there is no common solution region which can form feasible region.
Hence maximum z = 4x + 5y does not exists.



DIET PROBLEM

Q. **A Dietician recommends having 6 units of vitamin E and 8 units of vitamin K in their daily diet.**
For this Ramakant consume two foods A and B. Food A contains 3 units / Kg of vitamin E and 2 units / Kg of vitamin K. Food B contains 2 units / Kg of vitamin E and 4 units / Kg of vitamin K.
If costs to Ramakant 40 ₹ per Kg to purchase Food-A and 60 ₹ per Kg to purchase Food-B.
Formulate a mathematical model of LPP to minimize the expenditure of Ramakant

Sol. Let Ramakant purchase x kg and y kg of food A and food B respectively.

Food	Vitamin E	Vitamin K	cost
A	3	2	40
B	2	4	60
Quantity recommends by a dietician	6	8	

Constraints

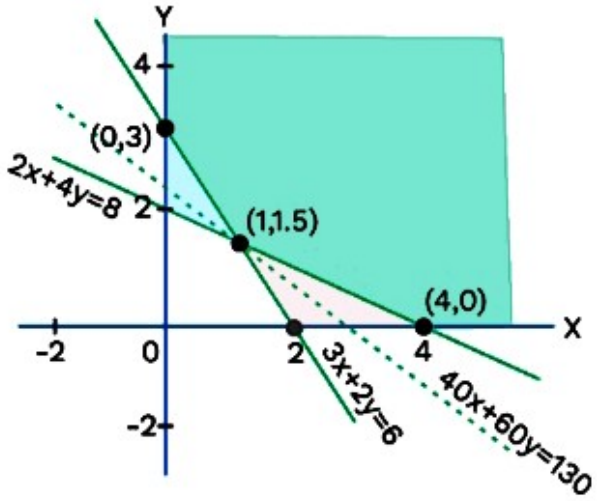
$3x + 2y \geq 6$

$2x + 4y \geq 8$

$x \geq 0, y \geq 0$

Objective Function: $Z = (40x + 60y)$

The feasible region determined by the given constrain is **unbounded** and corner points are found by solving linear equations.



Now evaluate the value of Z at corner points

Vertex point	Z = 40x + 60y
(0, 3)	180 = max
(1, 1.5)	130 = min
(4, 0)	160

Now draw the graph of $40x + 60y < 130$ as the region is unbounded and check whether the half-plane has any point common with the feasible solution. It is observed that there is no point common. So, 130 is the minimum value of objective function Z.

MANUFACTURING PROBLEMS :

Q. **Elon musk has 3 machines in the Space-X manufacturing factory, named M-1, M-2 and M-3. To produce rocket MARS-1 and MOON-2 all 3 machines are used M-1 and M-2 have the capacity to run at most 12 hrs. and M-3 must run for at least five hours a day. Time taken by each machine to produce one unit of each item is given in the table**

Item	No. of hours required		
	M-1	M-2	M-3
MARS-1	1	2	1
MOON-2	2	1	1.25

Elon can earn a profit of ₹ 600 and ₹ 400 on items MARS-1 and MOON-2 respectively. Find the optimal quantity of each rocket to maximize the profit and maximum profit?

Sol. Let x units of MARS-1 and y units of MOON-2 are produced.

Objective function Z = 600x + 400y

Constraints

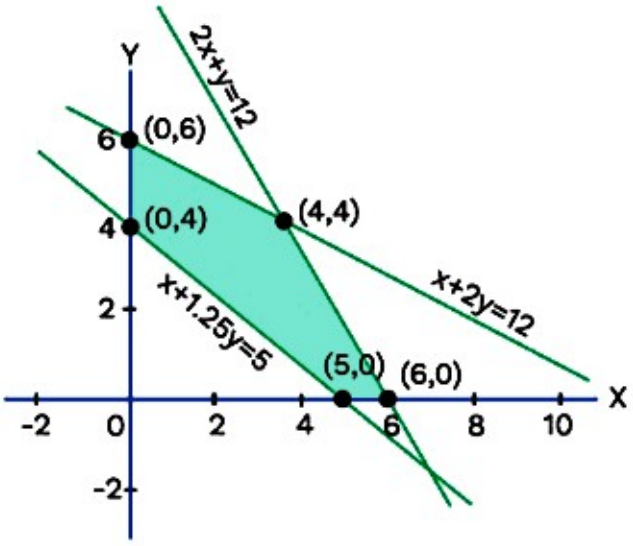
$x + 2y \le 12$

$2x + y \le 12$

$x + 1.25y \ge 5$

$x \ge 0, y \ge 0$

the feasible region is bounded as shown in the figure and corner points are determined by solving constrained equations.



Now evaluate the value of Z at corner points

Corner Points	Z = 600X + 400Y
(0, 6)	2400
(6, 0)	3600
(5, 0)	3000
(4, 4)	4000 = max
(0, 4)	1600 = min

Max profit to Elon is ₹ 4000 and the optimal number of each rocket is 4.

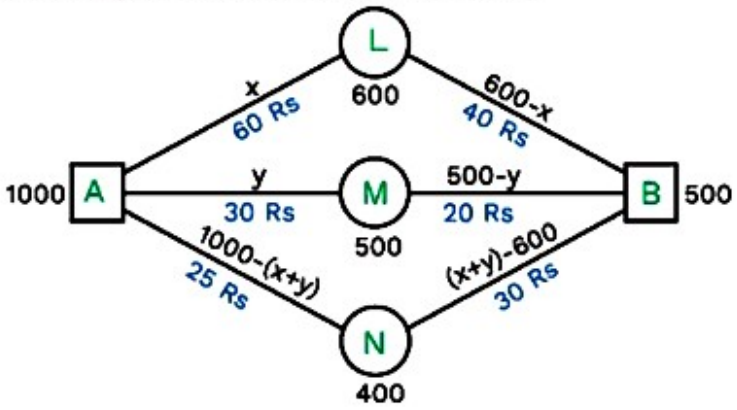
TRANSPORTATION PROBLEM:

Q. **Government of INDIA decides to distribute free ration for which the government stored food in godowns A and B of the capacity of 1000 quintals and 500 quintals respectively. Government supply ration to three ration shop L, M, and N whose requirements are 600,500 and 400 respectively. Transportation costs from godowns to ration shops are given in the following table.**

Transportation cost per quintal (in Rs)		
From/To	**A**	**B**
L	**60**	**40**
M	**30**	**20**
N	**25**	**30**

Indian government wants to minimize the cost of transportation to continuously supply free ration. What is the minimum cost?

Sol. Let Godown A supply x quintals and y quintals to L and M respectively.
For better understanding, we will draw a flow chart.



Constraints

$x \geq 0$	$600 - x \geq 0$	$500 - y \geq 0$	$x + y - 600 \geq 0$	$1000 - (x + y) \geq 0$
$y \geq 0$	$\Rightarrow x \leq 600$	$\Rightarrow y \leq 500$	$\Rightarrow x + y \geq 600$	$\Rightarrow x + y \leq 1000$

Objective Function:

Total expenditure on transportation

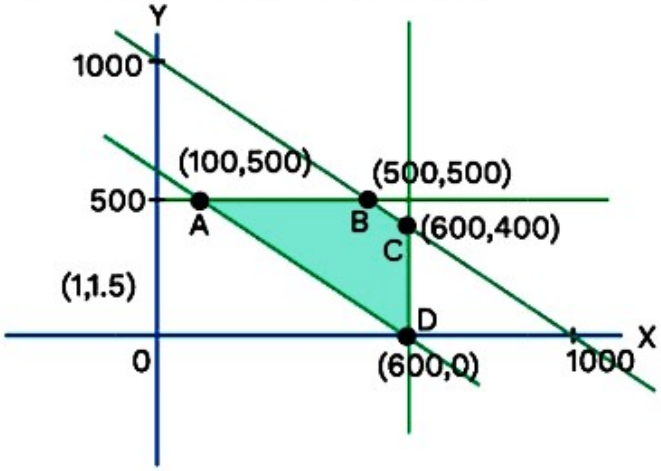
$Z = 60x + 30y + [1000 - (x + y)]25 + (600 - x)40 + (500 - y)20 + [x + y - 600]30$

$Z = 25x + 15y + 41000$

Corner points:

Corner points are determined by solving constraints equations

A(100, 500), B(500, 500), C(600, 400) and D(600, 0)

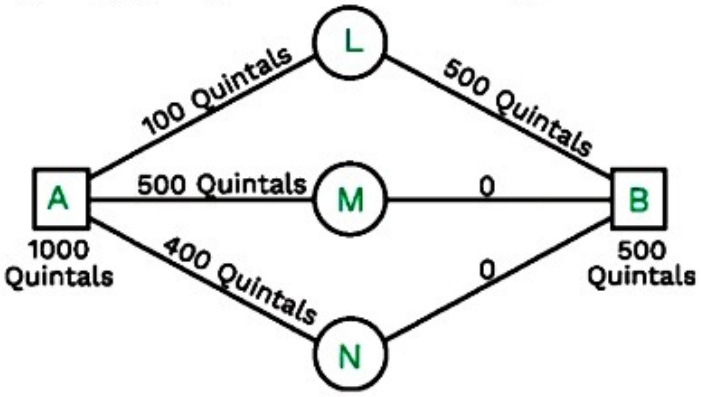


Now evaluate the value of Z at corner points

Vertex Point	Objective Function Z = 25x + 15y + 41000
(100, 500)	51000 = min
(500, 500)	61000
(600, 400)	62000 = max
(600, 0)	56000

Hence minimum expenditure on transportation is ₹ 51000

And optimal quantity supply is given in the flow diagram.



SUMMARY



- A linear Programming problem used to find the optimal value of objective function under the constrain of linear in-equalities.
- LPP model used for
  (i) Diet Problems
  (ii) manufacturing Problems
  (iii) Transportation Problems
- The solution region is common region represented by all linear constraints.
- Optimal solution is a point or set of points in solution region which maximize or minimize the objective function.
- For un-bounded region, maximum or minimum may not exist. However, if it exists, it must occur at vertex point.
- If two corner points of the solution region both give same optimal solutions. then every point on one segment joining these points is also an optimal solution.